

Exam Date & Time: 17-Jun-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MIT BLR - BTech VI Semester Makeup Examination, June -2025

BUSINESS INTELLIGENCE AND ANALYTICS [CSE-4018]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) A pharmaceutical company wants to improve its decision-making process related to drug inventory, distribution logistics, and market analysis. To achieve this, it plans to implement a Decision Support System (DSS) as part of its Business Intelligence (BI) strategy. Explain the help of Decision support system in informed decision-making for a pharmaceutical company with its advantages of integrating DSS with BI tools for this use case. (5)
 - A)
 - B)
- 2) A logistics company is facing challenges due to rising fuel costs, fluctuating customer demands, and increased competition. The management is considering implementing a Computerized Decision Support System (DSS) to adapt to these changing business conditions. Explain how changing business environments affect the company's decision-making and role of a computerized DSS in such a dynamic business setting. (5)
 - A)
 - B)
- 3) A retail company is facing inconsistent sales across regions. The analytics team is asked to uncover insights to support decision-making. Explain the cycle of Business Intelligence analysis from analysis till reports and how each stage helps solve the company's sales challenges. (5)
 - A)
 - B)
- 4) A company launched a new product based purely on market data without analysing trends or drawing insights from past campaigns. The product failed to gain transaction. In this failure context examine the role of data transformation into knowledge play and mention how the better use of the data-information-knowledge hierarchy have changed the outcome. (5)
 - A)
 - B)
- 5) A company is facing declining sales and wants to make informed decisions to improve its performance. Explain how can data-driven decision-making processes be applied in this scenario and how Business Intelligence tools can support effective business decision-making. (5)
 - A)
 - B)
- 6) Event-driven alerts are a critical style of Business Intelligence that enable real-time decision-making. Explain how event-driven alerts function within a BI system, and evaluate their impact on business operations by comparing them with other BI delivery styles such as dashboards and reports. (5)
 - A)
 - B)
- 7) Assume the retail company xyz faces challenges in integrating data from multiple store locations and ensuring real-time analytics. So as a BI analyst list the four key benefits of implementing BI system as solutions, major challenges the company may face during BI (5)
 - A)
 - B)

implementation and suggest a possible solution.

- A)
 - B) A financial services company is implementing a Business Intelligence (BI) system to analyse customer transactions and improve decision-making. However, the company must ensure that its BI implementation complies with legal and ethical requirements related to data privacy and security. (5)
Identify two major legal issues that the company may face while collecting and analysing customer data. Also suggest one measure the company can take to ensure compliance with legal and ethical standards in BI implementation.
- 5) Assume you are analysing log data from millions of devices for a smart city project. Explain the Hadoop ecosystem components you would use with a detailed diagram and its description in data storage and parallel processing. (5)
- A)
 - B) A large hospital system wants to reduce emergency room overcrowding by predicting which patients are likely to return within 30 days of discharge, based on electronic health records (EHR), lab results, and wearable device data. Survey the types of analytics to be applied at different stages of analysis ensuring data quality and veracity, also the tools or models to be employed. (5)

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